

SS-31

Mitochondria-Targeted Antioxidant | 10mg & 50mg x 10 Vials

Mitochondrial Protection | Cardioprotection | Renal Protection

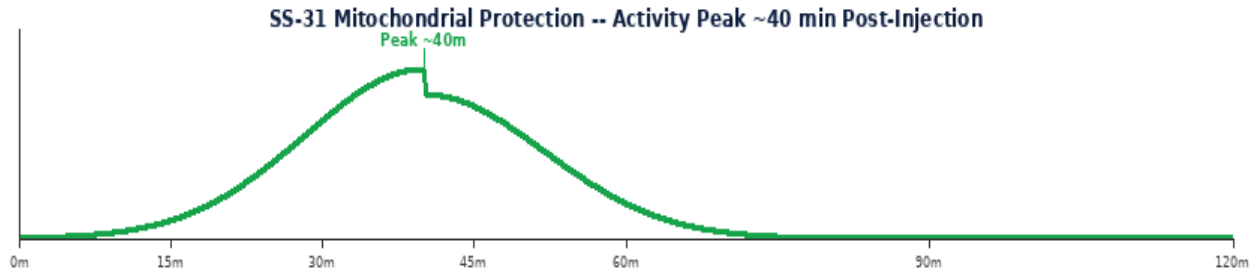
Purity: 99.8% | Endotoxin-Free | Research Use Only

SS-31 (Elamipretide, MTP-131) is a mitochondria-targeted tetrapeptide (D-Arg-2'6'-Dmt-Lys-Phe-NH2) that selectively concentrates in the inner mitochondrial membrane and stabilises cardiolipin. It is the most advanced mitochondria-targeted antioxidant in clinical development. Research applications: cardiac failure, renal protection, aging, and mitochondrial disease.



Parameter	Specification
Class	Mitochondria-targeted tetrapeptide antioxidant -- cardiolipin stabiliser
Vial Sizes	10mg x 10 50mg x 10 Purity 99.8%
Reconstitution	1 mL BAC Water: 10mg->10mg/mL 50mg->50mg/mL
Standard Dose	1,000-10,000 mcg (1-10mg) per injection SubQ
Frequency	1x daily SubQ no fasting required
Key Effects	Cardiolipin stabilisation, mitochondrial cristae preservation, ROS reduction, ATP restoration
Cycle & Storage	Ongoing with monitoring 2-8 deg C use within 28 days

MECHANISM OF ACTION & RESEARCH BENEFITS



Pathway	Mechanism	Benefit
Cardiolipin Stabilisation	Selectively targets inner mitochondrial membrane; binds and stabilises cardiolipin (essential phospholipid for cristae architecture); prevents cardiolipin peroxidation by cytochrome c; preserves mitochondrial cristae structure and ETC integrity	Mitochondrial architecture preservation; ETC efficiency; cardioprotection
ROS & ATP	Reduces superoxide and H ₂ O ₂ production from complexes I and III; restores mitochondrial membrane potential; improves ATP production in aged/diseased mitochondria; reduces oxidative damage cascades	Energy restoration; anti-aging; mitochondrial disease research
Cardiac & Renal	Reduces ischaemia-reperfusion injury in heart; preserves glomerular filtration; protects renal tubular cells from oxidative damage; clinical trials for heart failure (STEMI, chronic HF) and CKD	Cardiac protection; renal preservation; ischaemia research

Benefit	Context
Clinical Validation	SS-31/Elamipretide has completed Phase II clinical trials for heart failure and renal protection -- among the most clinically advanced mitochondria-targeted peptides.
Mitochondrial Architecture	Unique mechanism: targets cardiolipin, the structural lipid of the inner mitochondrial membrane -- restores cristae and ETC function.
Anti-Aging Research	Mitochondrial dysfunction is central to the aging process. SS-31 reverses functional decline in aged tissues in animal models.

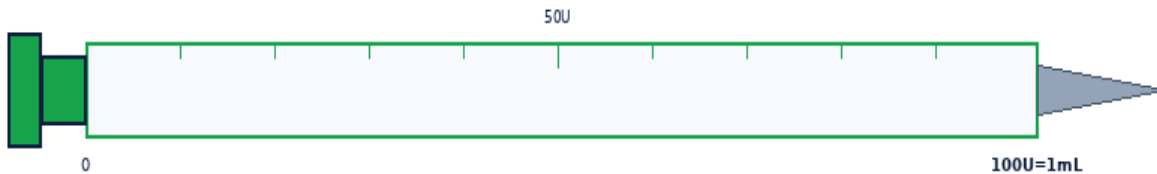
RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



1mL Insulin Syringe | 29-31G | SubQ

Formula: $\text{Vol(mL)} = \frac{\text{Dose(mcg)}}{\text{Conc(mcg/mL)}}$ | **Units:** $\text{Vol} \times 100$

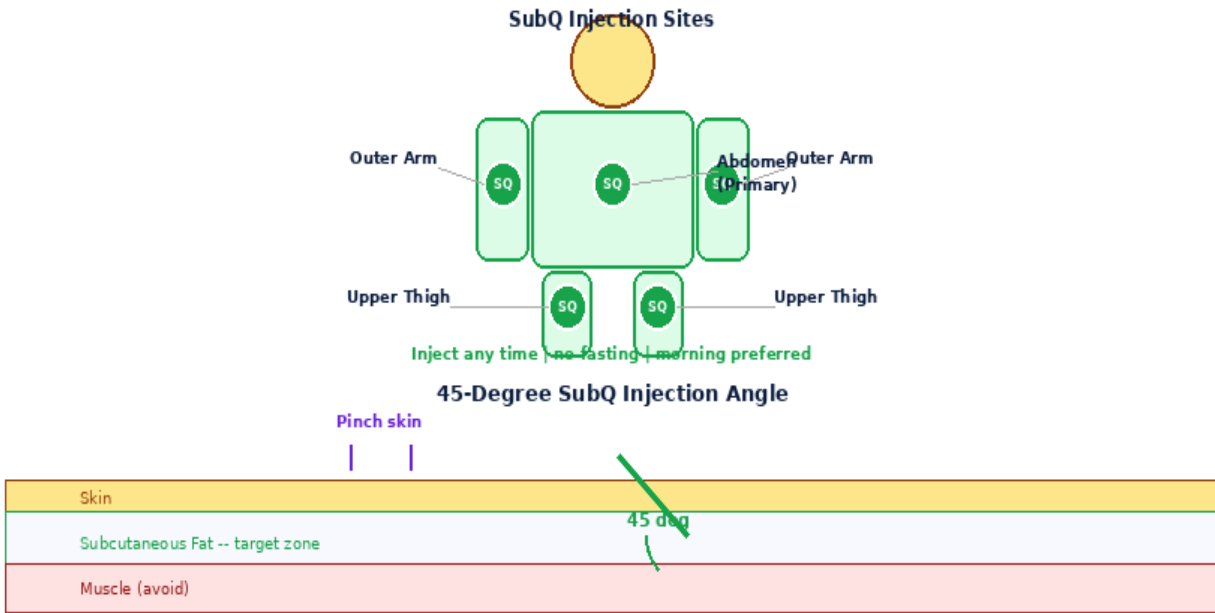
10mg x 10 -- 1mL BAC -> 10.0mg/mL (10,000mcg/mL)

Dose(mcg)	1000mcg	2500mcg	5000mcg	10000mcg
Vol(mL)	0.1000	0.2500	0.5000	1.0000
Units	10.0U	25.0U	50.0U	100.0U
Doses/Vial	10	4	2	1

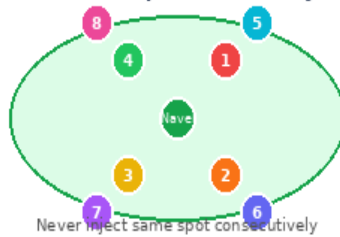
50mg x 10 -- 1mL BAC -> 50.0mg/mL (50,000mcg/mL)

Dose(mcg)	1000mcg	2500mcg	5000mcg	10000mcg
Vol(mL)	0.0200	0.0500	0.1000	0.2000
Units	2.0U	5.0U	10.0U	20.0U
Doses/Vial	50	20	10	5

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES



8-Point Rotation Map -- Rotate Every Injection



Step	Action	Detail
1	Wash & prepare	Wash hands. 29-31G syringe. No fasting required.
2	Draw dose	Draw volume per dose table. Standard 1-5mg.
3	Inject SubQ	Abdomen preferred. Pinch 2-3cm. 45 deg. Inject slowly.
4	Rotate & dispose	8-point rotation. Sharps container.

Protocol	Dose	Frequency	Duration	Notes
Standard	1,000-5,000 mcg	1x daily	Ongoing w/ monitoring	Morning injection preferred.
High-Dose Research	5,000-10,000 mcg	1x daily	8-12 wk	Clinical trial range. Monitor response.

Side Effect	Likelihood	Management
Injection site reaction	Common	Rotate sites. Fresh needle.
Fatigue (initial)	Occasional	Mitochondrial adaptation. Resolves in 1-2 weeks.
Nausea (high dose)	Occasional	Reduce dose. Inject with food.

Contraindications: Cancer history, pregnancy, under 18.

1. 10mg: 1mL BAC->10mg/mL 50mg: 1mL->50mg/mL.
2. No fasting required. Inject any time -- morning preferred.
3. Start at 1,000-2,500 mcg and escalate based on tolerance.
4. Unique mechanism: targets cardiolipin in inner mitochondrial membrane.
5. Most clinically advanced mitochondria-targeted peptide (Elamipretide -- Phase II trials).
6. Rotate injection sites through 8-point abdomen map.
7. Store 2-8 deg C. Use within 28 days. Never freeze.
8. Monitor cardiac and renal function on long protocols.
9. Contraindicated in cancer history. Not for pregnancy or under 18.

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